

WORKED EXAMPLE

Clip a gradient before using it in an update

Lecture 5 page 161 gives a ceiling on an individual positive derivative.

$$g = (6, -8), \quad c = 5, \quad w = (1, 1), \quad \eta = 0.1$$

g is the gradient vector supplied by backpropagation; w is the parameter vector. c is our threshold, called θ on the slide. We compare different clipping rules on the same original g .

$$w^{new} = w - \eta g_{used}$$

First choose and compute the clipped gradient. Then apply the parameter update. Clipping gradients does not mean clipping the values of the weights.

The symmetric coordinate and global-norm rules below are labeled supporting extensions to the slide's one-sided ceiling.

WORKED EXAMPLE

Work the slide one-sided ceiling literally

Only a derivative greater than c is replaced by c .

$$g_i > c \implies g_i^{used} = c; \quad \text{otherwise } g_i^{used} = g_i$$

$$6 > 5 \implies g_1^{used} = 5$$

$$-8 > 5 \text{ is false} \implies g_2^{used} = -8$$

$$g^{used} = (5, -8)$$

$$w^{new} = (1 - 0.1(5), 1 - 0.1(-8)) = (0.5, 1.8)$$

A large negative derivative remains untouched. The next extension bounds both positive and negative components.

WORKED EXAMPLE

Symmetric coordinate clipping

Supporting extension: cap each component between $-c$ and $+c$.

$$g_i^{clip} = \max(-c, \min(g_i, c))$$

$$g_1^{clip} = \max(-5, \min(6, 5)) = \max(-5, 5) = 5$$

$$g_2^{clip} = \max(-5, \min(-8, 5)) = \max(-5, -8) = -5$$

$$g^{clip} = (5, -5)$$

$$w^{new} = (1 - 0.1(5), 1 - 0.1(-5)) = (0.5, 1.5)$$

WORKED EXAMPLE

Global norm clipping: measure the vector length

Supporting extension: cap total Euclidean length instead of individual components.

$$\|g\|_2 = \sqrt{g_1^2 + g_2^2} = \sqrt{6^2 + (-8)^2} = \sqrt{100} = 10$$

The Euclidean norm is the length of the vector, calculated from its squared components. For a nonzero vector, choose one shared scale factor.

$$a = \min(1, c/\|g\|_2) = \min(1, 5/10) = 0.5$$

$$g^{clip} = ag = 0.5(6, -8) = (3, -4)$$

$$\|g^{clip}\|_2 = \sqrt{3^2 + (-4)^2} = 5$$

WORKED EXAMPLE

Use the norm-clipped vector in the update

Both components were scaled by the same factor .5.

$$w_1^{new} = 1 - 0.1(3) = 0.7$$

$$w_2^{new} = 1 - 0.1(-4) = 1.4$$

$$w^{new} = (0.7, 1.4)$$

$$\|\Delta w\|_2 = \eta \|g^{clip}\|_2 = 0.1(5) = 0.5$$

For this plain gradient-descent update, a global clipped-gradient length at most c gives a displacement length at most $\eta \cdot c$. It is not a bound on the parameter vector itself.

WORKED EXAMPLE

A component cap is not a length cap

The coordinate-clipped vector still has length greater than 5.

$$\|(5, -5)\|_2 = \sqrt{25 + 25} = \sqrt{50} \approx 7.071 > 5$$

$$\text{global ratios : } 3/6 = (-4)/(-8) = 0.5$$

$$\text{coordinate ratios : } 5/6 \neq (-5)/(-8)$$

Global clipping multiplies a nonzero vector by one positive number, preserving direction.

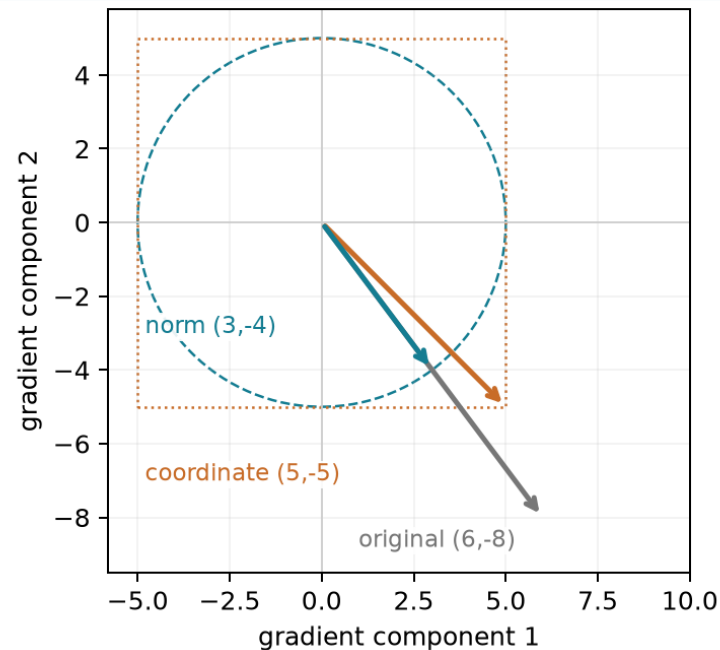
Coordinate clipping can use different effective factors and change direction.

A threshold 5 means different things in these two rules: largest allowed component magnitude versus largest allowed total vector length.

WORKED EXAMPLE

See the directions and allowed regions

Arrows are gradients from the origin, not parameter paths.



The square is the coordinate constraint; the circle is the norm constraint. The norm-clipped arrow lies on the same ray as the original vector.

WORKED EXAMPLE

Compare the resulting parameter states

Restart from $w=(1,1)$ for each independent comparison.

rule	gradient used	new parameters
none	(6,-8)	(.4,1.8)
slide positive ceiling	(5,-8)	(.5,1.8)
symmetric coordinate	(5,-5)	(.5,1.5)
global norm	(3,-4)	(.7,1.4)

The differences come from different rules, not arithmetic disagreements. Name the rule before computing it.

These supplied gradients illustrate updates; without a declared objective they do not establish which update achieves the lowest loss.

WORKED EXAMPLE

Handle a small vector, a boundary and zero

A clipping operation should not amplify a vector already within its cap.

$$g = (1, -2), \quad c = 5: \quad \|g\|_2 = \sqrt{5} < 5 \implies a = 1$$

$$g = (3, 4), \quad c = 5: \quad \|g\|_2 = 5 \implies a = 1$$

$$g = (0, 0) \implies g^{clip} = (0, 0)$$

For the zero vector, leave it unchanged directly; do not evaluate $c/0$. At the exact norm boundary, the shared scale is 1. Symmetric coordinate clipping also leaves all these example components unchanged.

INDEPENDENT PRACTICE

Your turn: a large negative component

Start from zero weights and compare all three rules.

$$g = (-12, 5), \quad c = 6, \quad w = (0, 0), \quad \eta = 0.1$$

Calculate the slide one-sided ceiling, symmetric coordinate clipping and global-norm clipping. Show both components, the norm and shared scale where needed, then each parameter update.

Explain why a component magnitude limit 6 does not imply a total gradient length limit 6.

WORKED ANSWER

Answer: one-sided and symmetric coordinate rules

The slide ceiling only changes components greater than +6.

$$-12 \leq 6, \quad 5 \leq 6 \implies g_{slide} = (-12, 5)$$

$$w_{slide}^{new} = (0 - 0.1(-12), 0 - 0.1(5)) = (1.2, -0.5)$$

$$g_1^{coord} = \max(-6, \min(-12, 6)) = -6$$

$$g_2^{coord} = \max(-6, \min(5, 6)) = 5$$

$$w_{coord}^{new} = (0 - 0.1(-6), 0 - 0.1(5)) = (0.6, -0.5)$$

WORKED ANSWER

Answer: norm, scale and clipped components

One shared factor rescales both coordinates.

$$\|g\|_2 = \sqrt{(-12)^2 + 5^2} = \sqrt{144 + 25} = 13$$

$$a = \min(1, 6/13) = 6/13$$

$$g_1^{norm} = (-12)(6/13) = -72/13 \approx -5.538462$$

$$g_2^{norm} = 5(6/13) = 30/13 \approx 2.307692$$

$$\|g^{norm}\|_2 = (6/13)(13) = 6$$

WORKED ANSWER

Answer: the final norm-clipped update

Subtracting a negative first component increases the first weight.

$$w_1^{new} = 0 - 0.1(-72/13) = 36/65 \approx 0.553846$$

$$w_2^{new} = 0 - 0.1(30/13) = -3/13 \approx -0.230769$$

$$\|(-6, 5)\|_2 = \sqrt{36 + 25} = \sqrt{61} \approx 7.81025 > 6$$

The last line checks the symmetric coordinate result: each magnitude is at most 6, but total length exceeds 6. Global clipping caps the length itself.

Common mistakes: ignoring a large negative component when a symmetric rule was requested, clipping weights instead of gradients, applying a different norm scale to each coordinate, or dividing by a zero norm.